2021 **BORED OF STUDIES TRIAL EXAMINATION**

2nd November

Mathematics Extension 2

General instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using a black or blue pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks: 100

Section I – 10 marks (pages 2–4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 5–13)

- Attempt Questions 11–16
- Allow about 2 hour and 45 minutes for this section

Section I

10 marks
Attempt Questions 1–10
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1−10.

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1 The complex number z lies on the fourth quadrant of the complex plane. If $\text{Arg}(z) = \theta$, which of the following represents an argument of $-z$?

(A) $\pi - \theta$

(B) $\pi + \theta$

(C) θ

(D) $-\theta$

2 The line ℓ has the vector equation

$$\underset{\sim}{r} = \begin{pmatrix} -1 \\ 4 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Consider an interval on the line ℓ where $0 \le \mu \le 1$.

What is the shortest distance from this interval to the point $\begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}$?

(A) $2\sqrt{2}$

(B) $2\sqrt{3}$

(C) 3

(D) 8

3 Which of the following statements is true?

(A) $\forall x > 0, \exists y < 0 : y > \dfrac{1}{x}$

(B) $\forall x > 0, \exists y < 0 : xy > 1$

(C) $\forall x < 0, \exists y > 0 : y < \dfrac{1}{x}$

(D) $\forall x < 0, \exists y > 0 : xy < 1$

4 Suppose that $z = -1 + i\sqrt{3}$. Which of the following is $\text{Arg}(z^5)$?

(A) $-\dfrac{2\pi}{3}$

(B) $\dfrac{2\pi}{3}$

(C) $\dfrac{10\pi}{3}$

(D) $\dfrac{20\pi}{3}$

5 A particle moves in simple harmonic motion according to the equation $v^2 = f(x)$, where v is particle's velocity at a given displacement x. Suppose that $f(1) = 0, f(5) = 0$ and $f'(2) = 4$. What is the period of the particle's motion?

(A) π

(B) $\sqrt{2}\pi$

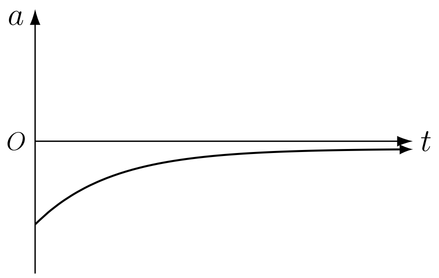
(C) 2π

(D) $2\sqrt{2}\pi$

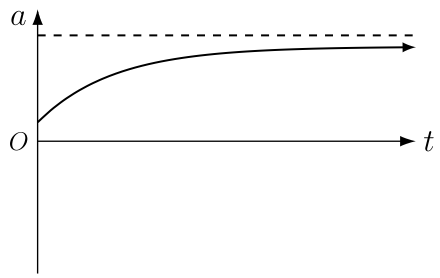
6 An object of mass m moves in a straight line and is subjected to a resistance force that is proportional to its speed. Define the direction of the object as the positive direction.

Let a be the object's acceleration at time t. Which of the following acceleration-time graphs best represents the motion of the object?

(A)



(B)



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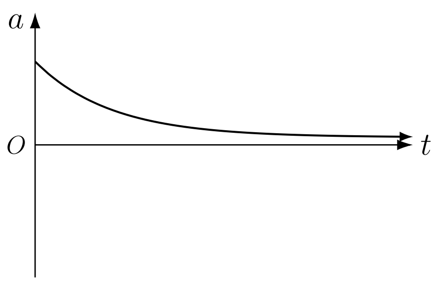
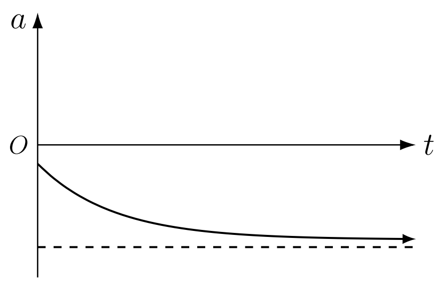


(D)



7 Let z be a complex number where $0 < \text{Arg}(z) < \dfrac{\pi}{4}$. Which of the following is correct?

(A) iz lies on the second quadrant and $z - iz$ lies on the first quadrant

(B) iz lies on the second quadrant and $z - iz$ lies on the fourth quadrant

(C) iz lies on the fourth quadrant and $z - iz$ lies on the first quadrant

(D) iz lies on the fourth quadrant and $z - iz$ lies on the second quadrant

8 Suppose that $f(x)$ is a continuous and differentiable function with only one turning point at $x = \epsilon$ over the interval (a, b), where $a < b$.

If $f(a) > f(b)$ and $f'(a) > f'(b)$, which of the following inequalities is always true?

(A) $f(\epsilon) \leq \dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx \leq f(a)$

(B) $f(a) \leq \dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx \leq f(\epsilon)$

(C) $f(b) \leq \dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx \leq f(\epsilon)$

(D) $f(\epsilon) \leq \dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx \leq f(b)$

9 Consider the proposition

“If p divides ab, then p divides a or p divides b”.

Which of the following examples disproves the contrapositive of the proposition?

(A) $p = 31,\ a = 402,\ b = 134$

(B) $p = 61,\ a = 194,\ b = 792$

(C) $p = 91,\ a = 539,\ b = 338$

(D) $p = 49,\ a = 199,\ b = 687$

10 Four points lie on a sphere, represented by the following vectors.

$$\underset{\sim}{a} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad \underset{\sim}{b} = \frac{1}{3}\begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}, \quad \underset{\sim}{c} = \frac{1}{3}\begin{pmatrix} 4 \\ 4 \\ 1 \end{pmatrix}, \quad \underset{\sim}{d} = \frac{1}{5}\begin{pmatrix} 2 \\ 11 \\ 0 \end{pmatrix}$$

Which of following vectors represents the centre of the sphere?

(A) $0\underset{\sim}{i} + 0\underset{\sim}{j} - \frac{2}{3}\underset{\sim}{k}$

(B) $0\underset{\sim}{i} + 0\underset{\sim}{j} - 2\underset{\sim}{k}$

(C) $0\underset{\sim}{i} + 0\underset{\sim}{j} + \frac{2}{3}\underset{\sim}{k}$

(D) $0\underset{\sim}{i} + 0\underset{\sim}{j} + 2\underset{\sim}{k}$

Section II

90 marks
Attempt Questions 11–16
Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet

(a) Use the substitution $u = \dfrac{1}{x}$ to evaluate $\displaystyle\int_{\frac{1}{2}}^{2} \frac{\ln x}{(x^2 - x + 1)^2}\, dx$. **4**

(b) Let a, b, x and y be non-zero real numbers such that

$$(x + iy)^2 = a + ib.$$

(i) Show that **3**

$$x^2 = \frac{\sqrt{a^2 + b^2} + a}{2} \quad \text{and} \quad y^2 = \frac{\sqrt{a^2 + b^2} - a}{2}.$$

(ii) Hence, explain why there are always two *distinct pairs* of values of (x, y). **1**

(c) An object of mass m is dropped from rest and is subjected to a gravitational force mg and a resistance force proportional to v^n, where g is the acceleration due to gravity, v is the velocity of the object at time t and n is a positive integer. Let V_T be the object's terminal velocity, for the given value of n.

(i) Show that when $n = 1$, then $v = V_T\left(1 - e^{-\frac{gt}{V_T}}\right)$. **2**

(ii) Find the velocity of the object when $n = 2$, in terms of g, t and V_T. **3**

(iii) Which scenario ($n = 1$ or $n = 2$) leads to the object approaching its own terminal velocity at a faster rate? Justify your answer using your results in parts (i) and (ii). **2**

End of Question 11

Question 12 (15 marks) Use the Question 12 Writing Booklet

(a) A child slides down a spiral shaped slide from height h to the ground. Let $\underset{\sim}{r}$ be the position vector of the child at time t seconds, relative to the centre of the slide at ground level. For a given positive constant a, the position vector is given by

$$\underset{\sim}{r} = (a \sin t)\underset{\sim}{i} + (a \cos t)\underset{\sim}{j} + \left(h - \frac{t^2}{4}\right)\underset{\sim}{k}$$

where $\underset{\sim}{i}$, $\underset{\sim}{j}$ are horizontal unit vectors and $\underset{\sim}{k}$ is a unit vector in the vertical direction.

(i) Find the time taken for the child to reach the ground in terms of h. **1**

(ii) When viewed directly from above, the child appears to be moving in a circular motion. Find the exact number of revolutions that the child appears to complete during the slide, in terms of h. **1**

(iii) Find the maximum speed of the child's motion in terms of a and h. **2**

(b) Suppose that for integer $n > 1$ **2**

$$P(z) = z^n \sin \alpha - z \sin n\alpha + \sin(n-1)\alpha,$$

where α is a real number not equal to a multiple of π.

Show that $z^2 - 2z\cos\alpha + 1$ is a factor of $P(z)$.

(c) Let P be a point represented by the position vector **2**

$$\underset{\sim}{p} = r\cos\theta\underset{\sim}{i} + r\sin\theta\underset{\sim}{j} + r\underset{\sim}{k},$$

where r and θ are parameters.

The point P also lies on a particular plane. This plane is perpendicular to the vector $r\cos\theta\underset{\sim}{i} + r\sin\theta\underset{\sim}{j} - r\underset{\sim}{k}$. Show that this plane also passes through the origin.

Question 12 continues on page 7

(d) Let a, b, c and d be positive real numbers. By considering the expansion of **2**

$$(a^2 - b^2)^2 + (c^2 - d^2)^2 + 2(ab - cd)^2,$$

or otherwise, show that

$$abcd \leq \left(\frac{a+b+c+d}{4}\right)^4.$$

(e) Let $P(x) = x^4 + px^3 + qx^2 + rx + s$ where p, q, r and s are real constants. $P(x)$ has real roots α, β, γ and δ. Consider the following statement.

If α, β, γ and δ are all positive, then $p^4 \geq 256s$.

(i) Using the result from part (d), or otherwise, show that this statement is always true. **1**

(ii) Write down the converse of this statement and explain (using a proof or counterexample) whether this converse is always true. **2**

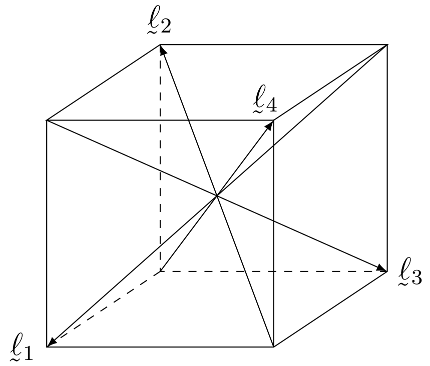
(iii) Write down the contrapositive of the statement. **1**

(iv) Hence, deduce that $x^4 - x^3 - 11x^2 + 9x + 18$ has at least one root that is not positive, given that all of its roots are real. **1**

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet

(a) Let $\underset{\sim}{\ell}_1, \underset{\sim}{\ell}_2, \underset{\sim}{\ell}_3$ and $\underset{\sim}{\ell}_4$ be vectors representing the diagonals that lie inside a cube, as shown in the diagram below. The vectors are placed such that the tips are equally spaced from each other.

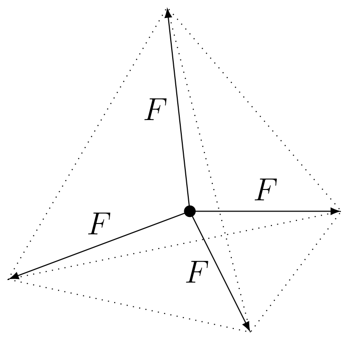


(i) Let $\hat{\underset{\sim}{x}}$ be a unit vector which makes an angle of α, β, γ and δ to the vectors $\underset{\sim}{\ell}_1, \underset{\sim}{\ell}_2, \underset{\sim}{\ell}_3$ and $\underset{\sim}{\ell}_4$ respectively. By considering dot products, show that **3**

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = \frac{4}{3}.$$

(ii) Hence, show that the obtuse angle between $\underset{\sim}{\ell}_1$ and $\underset{\sim}{\ell}_2$ is $\cos^{-1}\left(-\frac{1}{3}\right)$. **2**

(iii) A particle is subjected to four forces, each of magnitude F, in three dimensional space. The forces are directed such that the tips of their vectors form a regular triangular pyramid. **3**



By first considering projections in the direction of one of the force vectors, and using the result from part (ii), show that the net force on the particle is zero.

Question 13 continues on page 9

(b) An object of mass m is expected to move in simple harmonic motion about the origin with period $\frac{2\pi}{n}$ and amplitude a. It is initially at rest with a displacement of $-a$.

However, the object is instead subjected to both a resistance force of magnitude $mb\dot{x}$ and a positive driving force of magnitude $mc\sin(\omega t)$, where b, c and ω are non-negative constants, and $\dot{x}$ is the velocity of the object at time t.

(i) Find the object's displacement equation in terms of a and n, for $b = c = 0$. **1**

(ii) Now suppose that $b > 0$ and $c > 0$ instead. Explain why the object's acceleration equation is **1**

$$\ddot{x} = -n^2x - b\dot{x} + c\sin(\omega t).$$

(iii) When $t \geq \frac{2\pi}{\omega}$, the object reaches a steady state such that it moves in simple harmonic motion about the origin with period $\frac{2\pi}{\omega}$ and amplitude A. **3**

If the object is at $x = -A$ when $t = \frac{2\pi}{\omega}$, show that $\omega = n$.

(iv) Hence, deduce that $A = \dfrac{c}{bn}$. **1**

(v) Describe the effect on the object's motion if the magnitude of the resistance force is close to zero. **1**

End of Question 13

Question 14 (15 marks) Use the Question 14 Writing Booklet

(a) Given two distinct unit vectors $\hat{\underset{\sim}{m}}$ and $\hat{\underset{\sim}{n}}$, the two lines ℓ_1 and ℓ_2 have the vector equations $\underset{\sim}{r} = \lambda\hat{\underset{\sim}{m}}$ and $\underset{\sim}{r} = \mu\hat{\underset{\sim}{n}}$ respectively, for some parameters λ and μ.

Suppose that for some parameter θ,

$$\hat{\underset{\sim}{m}} = \left(\frac{\cos\theta + 1}{2}\right)\underset{\sim}{i} + \left(\frac{\sin\theta}{\sqrt{2}}\right)\underset{\sim}{j} + \left(\frac{\cos\theta - 1}{2}\right)\underset{\sim}{k}.$$

(i) Find a unit vector $\hat{\underset{\sim}{n}}$ that is independent of θ, such that the angle between ℓ_1 and ℓ_2 is always $\frac{\pi}{4}$. **2**

(ii) The line ℓ_2 intersects the plane $x - z = 4$ at the point C. Find the coordinates of C. **1**

(iii) The line ℓ_1 intersects the plane $x - z = 4$ at the point P. Show that as θ varies, P describes a circle of centre C and radius $2\sqrt{2}$. **3**

(b) Consider the polynomial

$$P(z) = z^3\left(\left(z + \frac{1}{z}\right)^3 + \left(z + \frac{1}{z}\right)^2 - 2\left(z + \frac{1}{z}\right) - 1\right).$$

(i) Show that **3**

$$P(z) = \left(z^2 - 2z\cos\frac{2\pi}{7} + 1\right)\left(z^2 - 2z\cos\frac{4\pi}{7} + 1\right)\left(z^2 - 2z\cos\frac{6\pi}{7} + 1\right).$$

(ii) Suppose that the roots of $x^3 + x^2 - 2x - 1$ are α, β and γ. Let **3**

$$A = \sqrt[3]{\alpha} + \sqrt[3]{\beta} + \sqrt[3]{\gamma}$$

$$B = \sqrt[3]{\alpha\beta} + \sqrt[3]{\beta\gamma} + \sqrt[3]{\gamma\alpha}.$$

Show that $A^3 = 3AB - 4$ and $B^3 = 3AB - 5$.

You may assume the following result without proof:

$$(u + v + w)^3 \equiv u^3 + v^3 + w^3 + 3(u + v)(v + w)(w + u).$$

(iii) Deduce that $AB = 3 - \sqrt[3]{7}$. **1**

(iv) Hence, find the exact value of $\sqrt[3]{\cos\frac{2\pi}{7}} + \sqrt[3]{\cos\frac{4\pi}{7}} + \sqrt[3]{\cos\frac{6\pi}{7}}$. **2**

End of Question 14

Question 15 (15 marks) Use the Question 15 Writing Booklet

(a) Suppose that for $0 < \alpha < \frac{\pi}{2}$, $0 < \beta < \frac{\pi}{2}$ and for integer $n \geq 0$

$$s_n = \sum_{k=0}^{n} e^{i(\alpha + k\beta)}.$$

Let P_n be a point on the complex plane represented by the complex number s_n.

(i) Using vectors, show that $\angle OP_0P_1 = \angle P_0P_1P_2$. **3**

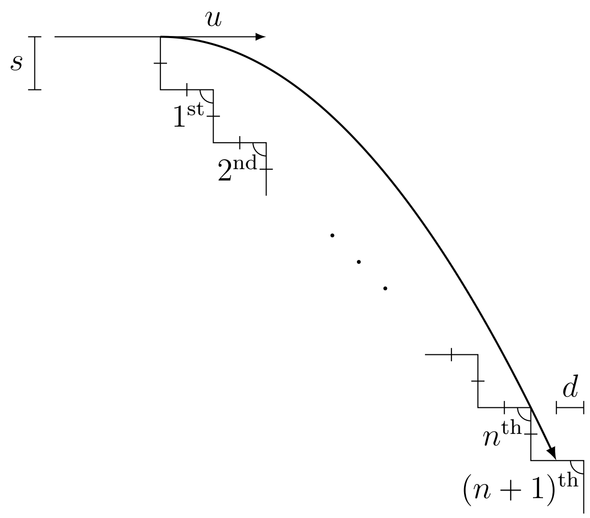
(ii) Suppose that the point P_{n+1} coincides with the origin. Find the maximum number of sides needed for the polygon $OP_0P_1P_2...P_n$, such there is only one possible value of the acute angle β. **2**

(b) Show that for $x \geq 0$ **3**

$$0 \leq \frac{x}{2} + 1 - \sqrt{1+x} \leq \frac{x^2}{8}.$$

Question 15 continues on page 12

(c) A particle is launched horizontally with an initial speed of u at the top of a flight of stairs. Each step has both a width and a height of s.



The particle just clears the edge of the nth step before landing on the $(n+1)$th step. Let d be the horizontal distance from the landing point to the edge of that step.

Let the initial position of the particle be the origin and $\underset{\sim}{r}$ be the displacement vector of the particle at time t given by

$$\underset{\sim}{r} = ut\underset{\sim}{i} - \frac{gt^2}{2}\underset{\sim}{j} \quad \text{(Do NOT prove this).}$$

(i) Show that $u = \sqrt{\dfrac{nsg}{2}}$. **2**

(ii) Show that $\dfrac{d}{s} = n + 1 - \sqrt{n(n+1)}$. **2**

(iii) Hence, use the inequality in part (b) to show that **1**

$$\frac{1}{2} \leq \frac{d}{s} \leq \frac{1}{2} + \frac{1}{8n}.$$

(iv) Show that when the particle is at the edge of the nth step, the vertical speed is twice the horizontal speed. **1**

(v) Describe the nature of the particle's *path* from the edge of the nth step to its landing point, when $n \to \infty$. **1**

End of Question 15

Question 16 (15 marks) Use the Question 16 Writing Booklet

(a) Let $I_n = \displaystyle\int_0^{\pi} \frac{\sin(nx)}{3 - 2\cos x}\, dx$ for some integer $n \geq 0$. Show that **2**

$$\sum_{k=1}^{2n+1} \left(\frac{3}{2} I_k - \frac{1}{2} I_{k+1} - \frac{1}{2} I_{k-1} \right) = 1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2n+1}.$$

(b) Suppose a, b, c and d are real numbers such that $a^2 + b^2 + c^2 + d^2 = 1$. Show that **2**

$$(1-a)(1-b) \geq cd.$$

(c) Let w be a fixed non-zero complex number and λ be a fixed real number such that $0 < \lambda < 1$. The complex number z satisfies the inequality

$$\left| \frac{z - \lambda w}{\lambda z - w} \right| < 1$$

where $z \neq \lambda w$ and $\lambda z \neq w$.

(i) Show that the complex number $\dfrac{z}{w}$ lies inside the unit circle. **2**

(ii) Hence, sketch the locus of z when $|w| = 1$. **1**

(d) Let n be a positive integer and λ be a real number such that $0 < \lambda < 1$. Let

$$P_n(z) = \sum_{k=0}^{n} \binom{n}{k} \lambda^{k(n-k)} z^k.$$

(i) Show that **2**

$$P_{n+1}(z) = z\lambda^n P_n\left(\frac{z}{\lambda}\right) + P_n(\lambda z).$$

(ii) Suppose that all the roots of $P_n(z)$ have a modulus of one and that $P_{n+1}(z)$ has a root of modulus less than one. **3**

Using the results from part (c), show that there exists a contradiction.

(iii) Hence, prove by mathematical induction that for all positive integers n, all the roots of $P_n(z)$ have a modulus of one. **3**

End of paper